publications. This quality-invariant influence of T5 publications persists when we restrict the sample to include faculty who published at least 4 or 5 journal articles during their first job spell (see Online Appendix Figures O-A16–O-A17).

The results presented in this section support the hypothesis that the T5 influence operates through channels that are independent of article quality. This finding is corroborated by responses to our survey of current tenure-track faculty at the top 50 U.S. economics departments. Junior faculty believe that there is at least a 0.89 probability that tenure committees will choose to tenure a candidate who possesses T5 publications over an identical candidate who possesses an identical number of non-T5 publications of the same quality. The existence of such a strong quality-independent influence of T5 publications suggests that tenure and promotion committees rely overwhelmingly on journal-level indicators of quality (T5 vs. non-T5) to predict the quality of individual articles. Such reliance on the T5 label is particularly problematic given the large intra-journal heterogeneity and inter-journal overlap in quality documented in Section 4 for articles published in T5 and non-T5 journals. The discussion in this section highlights the folly in relying on journal-level indicators of quality to predict individual article quality–it simultaneously generates errors in actual decision making and leads junior faculty to (correctly) believe that the T5 has a quality-independent effect on tenure decisions. The formation of such beliefs is likely to influence the direction of research for faculty who seek tenure and career advancement.

2.3 Duration Analysis of Time-to-Tenure

Table 1: Potential States of Employment for Untenured Tenure-Track Faculty in Period $t+1$ Relative to State in t

state = s	**Description**
0	Untenured tenure-track in the same T35 department as period t
1	Tenured in the same T35 department as period t
2	Untenured tenure-track in a different T35 department than period t
3	Not employed as a tenure-track faculty in a T35 department

This section expands on our analysis of the tenure–publication relationship by investigating the association between time-to-tenure and time-varying measures of publications in the four journal categories. We use a standard competing risks duration framework for the states given in Table 1. This section presents the multi-spell hazard specification used in this paper to estimate the duration relationships between tenure and T5 publications. The reader is referred to Text-Appendix Section 2 for a more detailed discussion of the model.

Consider a multi-spell model where each individual enters the post-PhD academic job market as an untenured assistant professor at one of the top 35 departments. The probability that an individual is employed in an untenured tenure-track position during the first period of any l^{th} spell of untenured tenure-track employment is 1. In subsequent periods, individuals can either remain untenured in the l^{th} spell of tenure-track employment ($s = 0$), begin a new spell $l + 1$ of untenured tenure-track employment in another T35 department ($s = 2$), or exit the sample by either receiving tenure within the department ($s = 1$) or by ceasing to be employed as a tenure-track faculty in a top 35 department ($s = 3$).[37]

Assuming that Weibull hazards generate survival times,[38] the hazard rate that an untenured tenure-track faculty in the l^{th} spell of employment transitions from state $s = 0$ to a new state $k \in \{1, 2, 3\}$ in time period t is parametrized by:

$$h^l_{0,k}(t) = \exp\left\{ \sum_{j \in \mathcal{J}} \left(\sum_{n=1}^{3} \alpha^{j,n}_{0,k} \cdot \mathbb{1}(j(t) \geq n) \right) + \mathbf{X}\boldsymbol{\beta}_{0,k} + \overline{\boldsymbol{C}}\boldsymbol{\eta}_{0,k} + \delta_{0,k}(l-1) + V^l_{0,k} \right\} t^{\gamma_{1,0,k}} \quad (1)$$

where $h^l_{0,k}(t)$ is the hazard rate of transitioning from state 0 to k in period-t of spell-l; $\mathbb{1}(j(t) \geq n)$ is an indicator for having n or more publications in journals of type $j \in \mathcal{J}$ as of time period t; $\mathcal{J} = \{T5, General, TierA, TierB\}$; $\mathbf{X}$ is a vector that includes fixed effects for authors' academic department as well as measures of observable characteristics including co-author characteristics, gender, quality of authors' PhD granting institution, years since

[37] Individuals cease to be employed as tenure-track faculty if they exit to a department below the top 35, move to an industry position, or transition to a non-tenure-track position in a top 35 department.

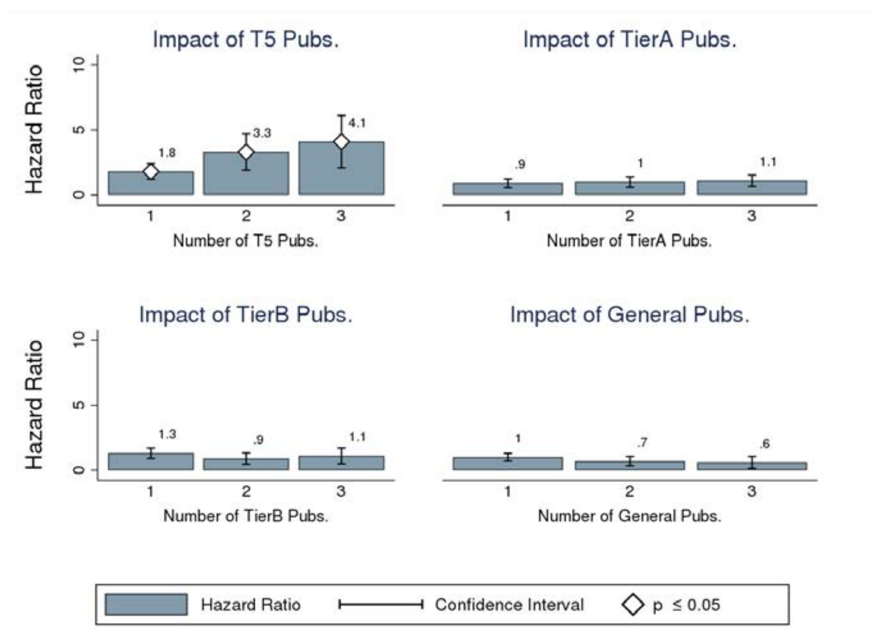
[38] see Text-Appendix Section 2 for a more detailed discussion of our duration model. The Text-Appendix presents the Weibull model as a special case of a duration model that employs a Box-Cox transformation.

graduation, and a control for total volume of publications ln(#Total Publications+1); $\overline{\boldsymbol{C}}$ is a vector of statistics that summarizes the distribution of field-adjusted citations received by each author[39]; $\gamma_{1,0,k}$ is the Weibull duration parameter; $\delta_{0,k}$ captures potential dependence between survival times and the number of spells that an individual has experienced prior to the current spell; and $V^l_{0,k} = C_{0,k}V$ is a one-factor spell-specific specification for individual-level unobserved heterogeneity.

The model imposes restrictions on the parameters associated with observed author characteristics and department fixed effects, forcing the parameters $\boldsymbol{\beta}$ to be equal across spells. We further restrict the parameters on the publication variables $\alpha^{j,n}_{0,k}$ to be constant across spells. This restriction is equivalent to assuming that tenure committees maintain the same publication standards for all untenured faculty regardless of the spell of employment.

2.3.1 Pooled Estimates of Hazard Rates and Time-to-Tenure

Figure 6: Relative Hazard Rates of Tenure Receipt Associated With Publications in Different Outlets

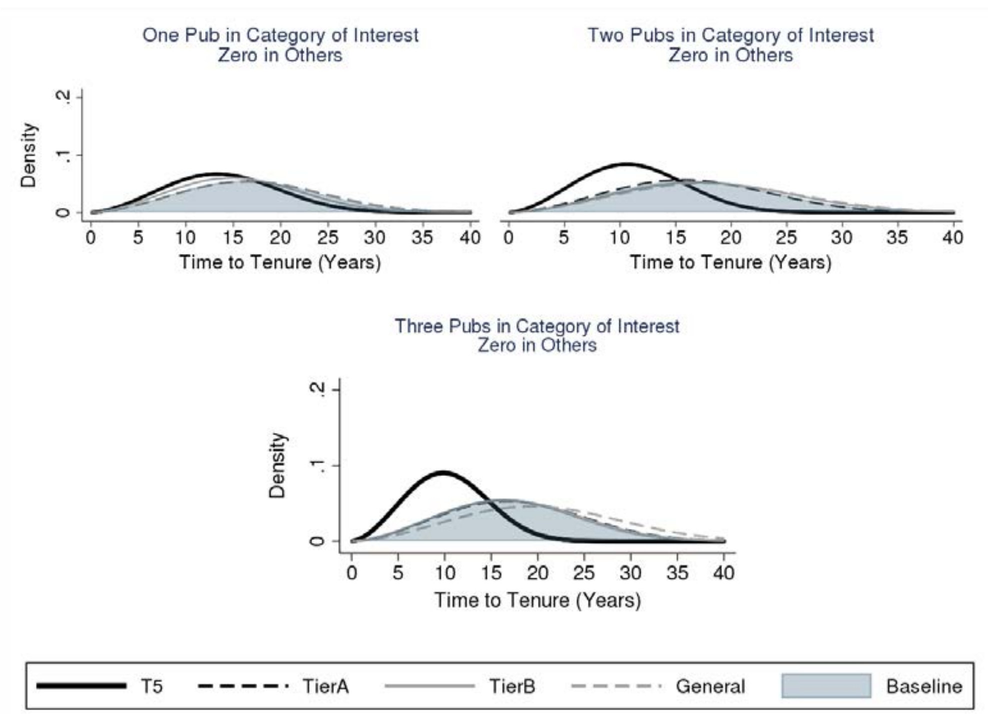


Note: This figure plots hazard ratios associated with different levels of publications in different outlets. Hazard ratios are obtained by estimating Text-Appendix Equation TA-13. White diamonds on the bars indicate that the prediction is statistically significantly different than 1 at the 5% level.

[39]see Footnote 29 for details

Figure 6 presents the increase in tenure hazards (rates of transition to tenure) associated with publishing different numbers of articles in the four journal categories. Estimates for individual parameters are presented in Online Appendix Table O-A17. The estimates show that the transition rates to tenure associated with individuals who publish two and three T5 publications are 3.3 and 4.1 times the transition rates associated with those who have never published in the T5. In comparison, the transition rates associated with those who have published three Tier A or Tier B publications is no higher than 1.1 times the hazards associated with individuals who have never published in these outlets. None of the estimates for the non-T5 hazard ratios are statistically significant at the 5% level.

Figure 7: Densities of time-to-tenure (Weibull Distribution)



Note: This figure plots distributions of time-to-tenure associated with different levels of publications in four different types of journals. Densities of time-to-tenure are derived from estimation of Equation TA-13. The blue shaded region in each plot represents the distribution of time-to-tenure associated with not having any publications in any journal.[40]

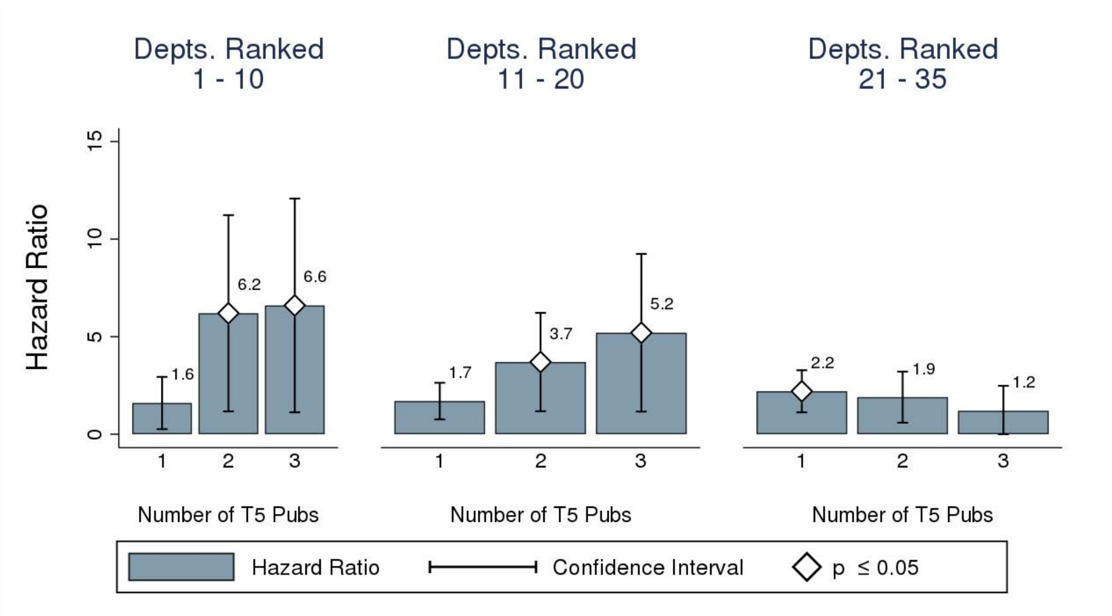
The differences in hazard rates translate into differences in the time required to attain tenure. Figure 7 plots predicted densities of time-to-tenure associated with publishing different numbers of articles in the four journal categories.[41] Publishing in the T5 is associated

[41]Each panel plots a baseline density associated with having no publications in any of the four journal

with large decreases in the expected time-to-tenure as indicated by the large leftward shift in the T5-specific density of predicted time-to-tenure. In comparison, publications in non-T5 journals are associated with negligible deviations from the baseline distribution.

2.3.2 Estimates of Hazard Rates by Department Rank

Figure 8: Relative Hazards of Tenure Associated With Different Levels of T5 Publications (By Department Rank)



Note: This figure plots department quality-specific hazard rates of tenure associated with different levels of publications in the T5. The department quality-specific hazard rates are estimated by interacting the publication parameters in Equation TA-13 with time-specific indicators for whether an author is hired by a department that belongs to each of the three department-quality groupings.

This section presents hazard estimates corresponding to three rank-based groupings of deparments: Top 10, Top 11–20, Top 21–35. To estimate rank-specific hazard ratios, we interact the publication variables in Equation 1 with indicators for being employed by a department in one of the three rank-based groups during period t.[42] The estimates are

categories. Journal category-specific densities are overlaid on this baseline density to highlight the deviation in time-to-tenure associated with publishing in the different categories. The first subfigure plots the densities associated with publishing one article in the journal category of interest, and none in the other three. The remaining two subfigures analogously plot densities associated with publishing two and three articles in the journal category of interest while holding the number of publications in the other three categories at zero.

[42] See Text-Appendix Section 2.3 for details.